

Q:01 Software Development Life Cycle (SDLC):



SDLC Lifecycle Diagram and Brief Explanation

The Software Development Life Cycle (SDLC) is a structured and systematic process used to develop high-quality software. It defines the major stages involved in transforming a software idea or requirement into a working system and maintaining it after deployment.

SDLC Workflow

Requirement Analysis → System Design → Development → Testing → Deployment
→ Maintenance → Feedback → New/Updated Requirements

The SDLC is a continuous and iterative process. After deployment, user feedback and maintenance activities may identify new requirements or improvements, which can start another development cycle.

1. Requirement Phase

In this phase, the customer and stakeholders' needs are collected, analyzed, documented, and validated. The development team determines what the software should accomplish and defines the functional and non-functional requirements.

Main activities: Requirement gathering, analysis, documentation, and validation.

2. Design Phase

The approved requirements are transformed into a technical blueprint for the software. The system architecture, database structure, user interface, modules, and technologies are planned.

Main activities: System architecture, database design, UI design, and technical design.

3. Build/Development Phase

In this phase, developers write the source code and implement the system according to the approved design. Individual modules and features are developed and integrated.

Main activities: Coding, module development, integration, and code review.

4. Testing Phase

The developed software is systematically tested to identify bugs, errors, security issues, and functional problems. The system is checked against the requirements to ensure that it works correctly.

Main activities: Functional testing, integration testing, system testing, and defect fixing.

5. Deployment/Delivery Phase

After successful testing and approval, the software is released and installed in the target environment so that users can access and use the system.

Main activities: Release preparation, installation, configuration, and user deployment.

6. Maintenance Phase

After deployment, the software is continuously monitored, maintained, and improved. Bugs are fixed, performance may be optimized, security updates are applied, and new features may be added based on user feedback.

Main activities: Bug fixing, updates, performance improvement, security maintenance, and enhancement.

SDLC Workflow Diagram

The given diagram represents the following cycle:

Requirement Phase

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Design Phase

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Build/Development Phase

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Testing Phase

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Deployment/Delivery Phase

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Maintenance Phase

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Feedback & New Requirements

↪ Back to Requirement Phase

Q:02

For this practical exercise, I selected **ChatGPT** as the AI tool.

ChatGPT was used as an assistant for:

- Generating Agile user stories
- Developing acceptance criteria
- Preparing a Sprint Backlog
- Identifying missing or unclear requirements
- Supporting requirement refinement

The generated output was considered an initial draft rather than a final specification. I manually reviewed and improved the output according to the expected requirements of a realistic E-Commerce application.

Five user stories for an E-commerce application

US-01: Customer Account Registration

As a new customer, I want to create an account using my personal information, so that I can securely access the E-Commerce platform and manage my orders.

Business Value:

Allows customers to create personalized accounts and provides a foundation for secure shopping and order management.

US-02: Product Search and Filtering

As a customer, I want to search and filter products by name, category, price, and availability, so that I can quickly find products that match my requirements.

Business Value:

Makes product discovery faster and improves the overall shopping experience

US-03: Shopping Cart Management

As a customer, I want to add, remove, and update products in my shopping cart, so that I can review and modify my selected items before checkout.

Business Value:

Gives customers flexibility and control over their purchases before placing an order.

US-04: Secure Online Payment

As a customer, I want to pay for my order using a secure online payment method, so that I can complete my purchase conveniently and safely.

Business Value:

Enables customers to complete transactions securely while providing a convenient checkout experience.

US-05: Order Tracking

As a customer, I want to track my order status, so that I can know whether my order is being processed, shipped, or delivered.

Business Value:

Improves transparency and increases customer confidence after an order is placed.

Selected User Story

For detailed acceptance criteria, **US-04: Secure Online Payment** has been selected because payment is a critical part of an E-Commerce system and requires clear functional validation.

Selected User Story

As a customer, I want to pay for my order using a secure online payment method, so that I can complete my purchase conveniently and safely.

Acceptance Criteria5. Acceptance Criteria

ID Acceptance Criteria

- AC-01 Given the customer has items in the cart, when the customer proceeds to check out, then the system shall display the order subtotal, delivery charge, and final payable amount.
- AC-02 Given the customer is on the payment page, when available payment options are displayed, then the customer shall be able to select a valid payment method.
- AC-03 Given the customer has selected a payment method, when the customer submits the payment information, then the system shall validate the required information before processing the transaction.
- AC-04 Given the payment information is valid, when the payment gateway successfully processes the transaction, then the system shall mark the payment as successful and confirm the order.
- AC-05 Given the payment transaction fails, when the payment gateway returns a failure response, then the system shall display an appropriate error message and allow the customer to retry.
- AC-06 Given a payment has failed, when the transaction is unsuccessful, then the system shall not confirm the order or mark it as paid.
- AC-07 Given the payment is successful, when the order is confirmed, then the system shall generate an order confirmation and update the customer's order status.

Acceptance Criteria Quality

These criteria are designed to be:

- **Specific** – Each criterion describes a clear expected behavior.
- **Measurable** – The result can be verified.
- **Testable** – Each criterion can be converted into a test case.
- **Relevant** – All criteria directly support the selected user story.
- **Complete** – Both successful and failed payment scenarios are considered.

. Sprint Backlog

ID	Sprint Backlog Task	User Story	Priority	Story Points	Status
T01	Design customer registration interface	US-01	High	3	To Do
T02	Implement registration and input validation	US-01	High	5	To Do
T03	Develop product search functionality	US-02	High	5	To Do
T04	Implement product category and price filters	US-02	Medium	3	To Do
T05	Develop shopping cart functionality	US-03	High	5	To Do
T06	Implement add, remove, and quantity update operations	US-03	High	3	To Do
T07	Integrate secure online payment functionality	US-04	High	8	To Do
T08	Implement payment validation and failure handling	US-04	High	5	To Do
T09	Develop order status and tracking functionality	US-05	Medium	5	To Do
T10	Test order tracking and status updates	US-05	Medium	3	To Do

What the AI Did Well

ChatGPT was particularly useful for the initial requirements generation and structuring.

It successfully:

- Generated relevant E-Commerce features.
- Used the standard Agile user story format.
- Identified important customer requirements.
- Suggested appropriate acceptance criteria.
- Created a structured Sprint Backlog.
- Helped identify functional requirements quickly.
- Reduced the time required for initial planning.
- Provided a useful starting point for further human analysis.

AI Output Review

After receiving the initial response from ChatGPT, I did not use the output without verification. I reviewed it from an Agile and Software Engineering perspective and identified areas where improvements were required.

1. Improved User Story Quality

The wording was refined to make each story follow a consistent Agile structure.

2. Added Business Value

Business value was added to explain why each feature is important to the customer and the system.

3. Improved Acceptance Criteria

The acceptance criteria were changed into Given-When-Then format to make them easier to test.

4. Added Negative Scenarios

Payment failure and retry scenarios were included because a real payment system must handle unsuccessful transactions.

5. Improved Sprint Planning

The Sprint Backlog was divided into smaller and manageable development tasks.

6. Added Estimation

Story points were included to provide a relative estimation of development effort.

7. Added Sprint Goal

A clear Sprint Goal was defined to explain what the development team intends to achieve during Sprint 1.

Why Human Review Is Necessary

Human review is essential because software requirements must reflect real business needs, stakeholder expectations, technical constraints, security requirements, and project scope.

AI can generate suggestions quickly, but it cannot independently guarantee that every requirement is correct or feasible for a particular project.

For example, an AI may suggest an online payment feature, but a human Business Analyst must determine:

- Which payment methods are supported.
- What security requirements apply.
- How failed transactions should be handled.
- What happens if money is deducted but the order is not confirmed.
- What information should be stored.
- Whether the feature is achievable within the sprint.

Therefore:

AI provides speed and assistance, while human expertise provides validation, context, judgment, and accountability.

Some screenshots of AI interaction

Act as a professional Agile Business Analyst with experience in Scrum and requirements engineering. For a realistic E-Commerce application, create five high-quality user stories using the standard format "As a [user], I want [goal], so that [benefit]." Give each user story a unique ID, title, and business value. Then select the most appropriate user story and create at least five clear, specific, measurable, and testable acceptance criteria using Given-When-Then format. Finally, create a Sprint 1 Backlog based on the five user stories. Include task ID, task description, related user story, priority, story points, and status. Make sure the tasks are realistic, properly broken down, and suitable for a university-level Software Engineering assignment. Keep the output structured, professional, and concise.

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AI-Generated Output – E-Commerce Application

1. Five User Stories

US-01: Customer Account Registration

3. User Stories

US-01: Customer Account Registration

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Business Value:

Enables customers to complete transactions securely while providing a convenient checkout experience.

US-05: Order Tracking

As a customer, I want to track my order status, so that I can know whether my order is being processed, shipped, or delivered.

Business Value:

give me a sprint backlog To develop the core E-Commerce shopping flow that allows customers to create an account, find products, manage their shopping cart, and complete a secure online purchase.

Absolutely. Since your **Sprint Goal** is:

"To develop the core E-Commerce shopping flow that allows customers to create an account, find products, manage their shopping cart, and complete a secure online purchase."